

Philipp Nazari

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[Google Scholar](https://scholar.google.com/citations?user=phnazari)

Education

PhD	Max Planck ETH Center for Learning and Systems , Computer Science - CLS PhD Fellow, advised by Konstantin Rusch and Prof. Fanny Yang - Research Areas: Efficient AI, Geometric Deep Learning, AI for Science	July 2025 – July 2028
MsC	ETH Zurich , Mathematics - GPA: 5.78 (Swiss grading system) - Focus Areas: Differential Geometry, Machine Learning	Sept 2023 – Feb 2025
BsC	Ruprecht-Karls-University Heidelberg , Mathematics - GPA: 1.3 (German grading system) - Focus areas: Differential Geometry, Analysis, Machine Learning	Oct 2019 – July 2023
	University of Bergen , Mathematics - Exchange Semester at the University of Bergen, Norway - Focus Areas: Algebraic Topology, Differential Geometry	Aug 2021 – Jan 2022
BsC	Ruprecht-Karls-University Heidelberg , Physics - GPA: 1.3 (German grading system) - Focus areas: Theoretical Physics, Machine Learning	Oct 2019 – Aug 2022

Experience

Heidelberg Collaboratory for Image Processing (HCI) , Research Assistant	Heidelberg, Germany Aug 2022 – Mar 2023
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Publications

On State Reduction in Linear Attention <i>Philipp Nazari</i> , T. Konstantin Rusch openreview.net/forum?id=KkN6enKAlU (<i>Oral</i> AdaptFM workshop at ICML 2026)	June 2026
The Key to State Reduction in Linear Attention: A Rank-based Perspective <i>Philipp Nazari</i> , T. Konstantin Rusch arxiv.org/abs/2602.04852 (Preprint)	February 2026
The Curious Case of In-Training Compression of State Space Models Makram Chahine, <i>Philipp Nazari</i> , Daniela Rus, T. Konstantin Rusch openreview.net/forum?id=LtzmeSMBTW (ICLR 2026)	October 2025
Geometric Autoencoders – What You See is What You Decode <i>Philipp Nazari</i> , Sebastian Damrich, Fred Hamprecht proceedings.mlr.press/v202/nazari23a.html (ICML 2023)	July 2023

Software

Linux: State Space Models in JAX <i>Philipp Nazari</i> , Francesco Ruscio, Benedict Armstrong github.com/camail-official/linux	October 2025
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Research Projects

Geometric Encoder Regularization in Autoencoders	Juli 2024
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- Semester Project at ETH with Prof. Thomas Hofmann. [Paper](#) and [code](#)

Entropy Aware Message Passing in Graph Neural Networks

Mai 2024

- Project spun out of the Deep Learning course at ETH by Prof. Thomas Hofmann. [Paper](#) and [code](#)

Talks

Guest Lecturer

November 2024

- Guest lecturer in the course "Machine Learning and Physics" at Ruprecht-Karls-University Heidelberg by Prof. Dr. Fred Hamprecht: "An Introduction to Autoencoders and (Geometric) Regularization Techniques"